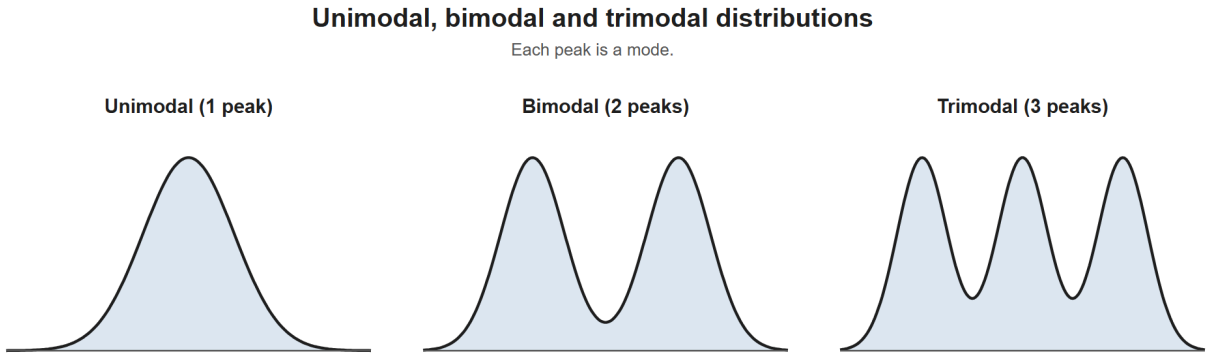
Lesson 4.5 - Skewness with Box Plots

> **Video:** Skewness - Right, Left & Symmetric Distribution - Mean, Median, & Mode With Boxplots by The Organic Chemistry Tutor **Chapter:** 4 - Measures of Position

Number of modes

The **mode** is the value with the highest frequency, which is the **peak** of the distribution.

Shape	Peaks	Name
One peak	1	**Unimodal**
Two peaks	2	**Bimodal**
Three peaks	3	**Trimodal** (or, in general, **multimodal**)



The number of peaks gives the number of modes.

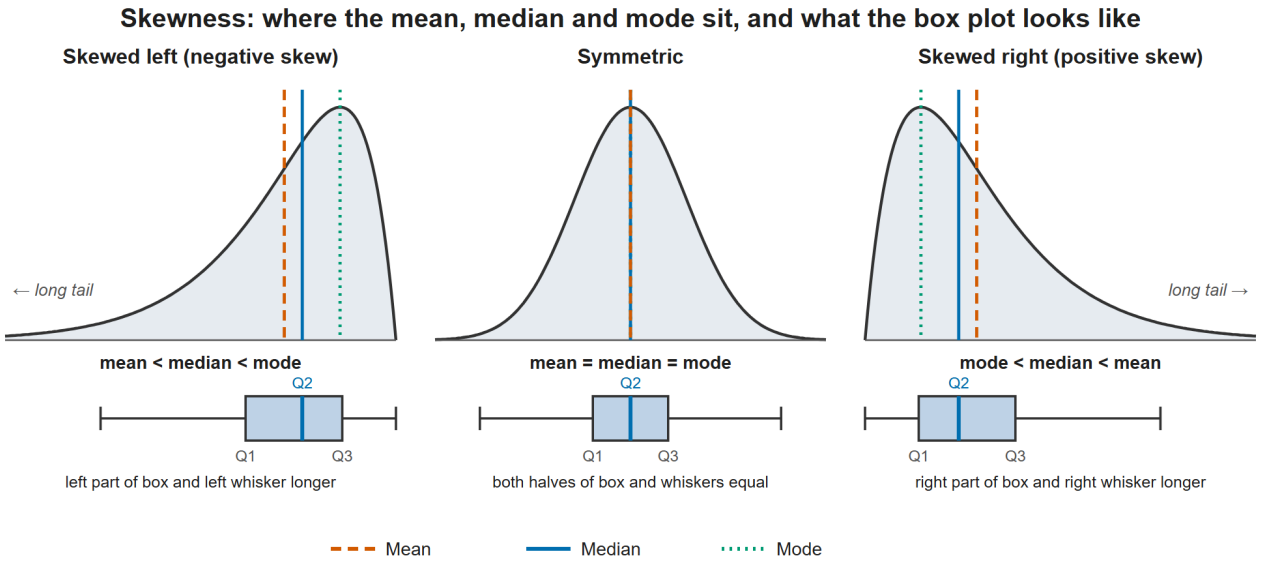
The three shapes (unimodal distributions)

Shape	Tail	Order of mean, median, mode
Symmetric	Left and right sides mirror each other	mean = median = mode
Skewed right (positive skew)	Long tail to the **right**	mode < median < **mean**
Skewed left (negative skew)	Long tail to the **left**	**mean** < median < mode

Reading left to right along the x-axis:

- **Skewed right:** peak (mode) → median → mean → long tail on the right
- **Skewed left:** long tail on the left → mean → median → peak (mode)

> **Memory aid:** the **mean is pulled toward the tail**. The mode stays at the peak, and the median sits between them.



Curves: Gamma(2, 1), a normal curve, and the mirror image of Gamma(2, 1). Whiskers drawn to the 1.5 × IQR fences.

Where the mean, median and mode fall for each shape, with the matching box plot underneath. The mean is always pulled toward the tail.

Reading skewness from a box plot

Let the left part of the box be $Q_2 - Q_1$ and the right part be $Q_3 - Q_2$.

Symmetric

```
|-------[=====|=====]-------|
```

- $Q_3 - Q_2 = Q_2 - Q_1$ (equal halves of the box)
- Whiskers of equal length

Skewed right (positive)

Any of these indicates right skew:

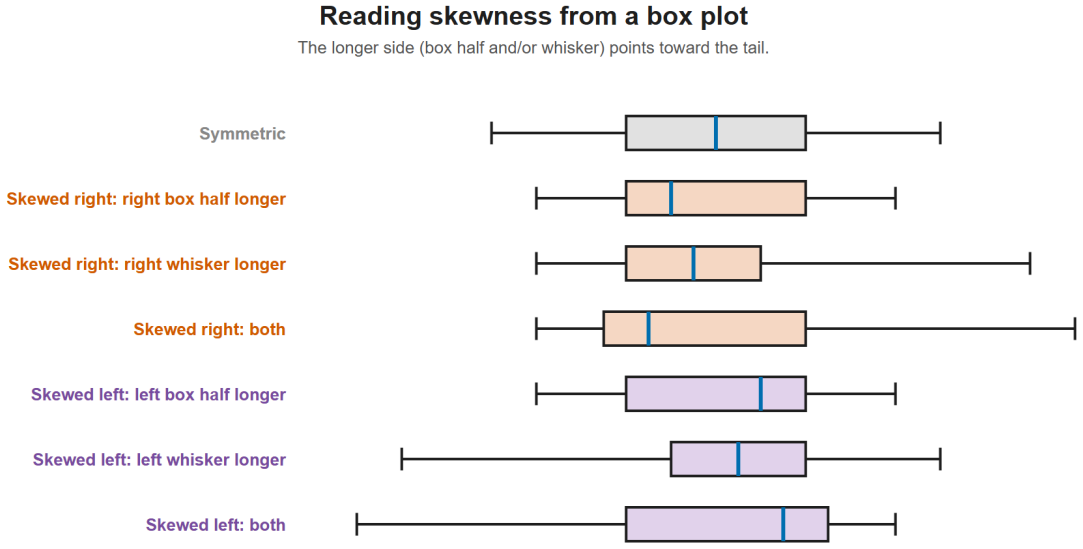
```
(a) right half of box longer:       |----[==|=======]----|
(b) equal box halves, longer right
    whisker:                        |--[====|====]----------|
(c) both:                           |--[==|========]------------|
```

- $Q_3 - Q_2 > Q_2 - Q_1$, and/or the right whisker is longer.

Skewed left (negative)

```
(a) left half of box longer:        |----[=======|==]----|
(b) equal box halves, longer left
    whisker:                        |----------[====|====]--|
(c) both:                           |------------[========|==]--|
```

- $Q_2 - Q_1 > Q_3 - Q_2$, and/or the left whisker is longer.



The three ways a box plot can show skew: a longer box half, a longer whisker, or both. The longer side always points toward the tail.

Key takeaways

- Unimodal, bimodal, multimodal = one, two, many peaks.
- Symmetric: mean = median = mode.
- Right skew: mean > median > mode. Left skew: mean < median < mode.
- On a box plot, the **longer side** (box half and/or whisker) points in the direction of the skew.